

Disproof of Conjecture 00000008579: the eigenvalue multiplicities of the Ornstein–Uhlenbeck operator are not the partition numbers

jilint777

2026-10-04

Abstract

Conjecture 00000008579 makes four claims about the Ornstein–Uhlenbeck operator on Wiener space: its spectrum is $\{0, 1, 2, \dots\}$; the multiplicity of the eigenvalue n is the partition number $p(n)$; the spectral counting function has Hardy–Ramanujan asymptotics; and its second-order correction is the Rademacher series. We refute the second claim (the “partition multiplicity law”), and hence the conjunction. The eigenvalue n of $N = -L$ has the n -th Wiener chaos as its eigenspace. On a Wiener space with infinitely many coordinates this space is infinite-dimensional for every $n \geq 1$: for example $W(e_0), W(e_1), \dots$ are linearly independent eigenvectors for the eigenvalue 1, while $p(1) = 1$. On the finite-dimensional Gaussian space $\mathbb{R}^d$ the multiplicity is $\binom{n+d-1}{d-1}$. For $d \geq 2$ this gives $\text{mult}(1) = d \geq 2 \neq p(1)$. For $d = 1$ every multiplicity is 1, which differs from $p(n)$ for every $n \geq 2$. So the law fails on every Gaussian space, at $n = 1$ or $n = 2$. The decisive objects are formalized in Lean 4 (core library only): polynomials, ∂_i , multiplication by x_i , the generator $L = \sum_i (\partial_i^2 - x_i \partial_i)$, eigenspaces, linear independence, multiplicity and $p(n)$. The final theorem refutes the law in every dimension $d \geq 0$ and on the infinite-dimensional (cylinder) Wiener space.

1 The conjecture and its precise reading

Definition: The Malliavin spectrum of the Ornstein–Uhlenbeck operator on Wiener space. *Conjecture:* The spectrum consists of the nonnegative integers (a Hermite spectrum law); the eigenvalue multiplicities are the partition numbers of the corresponding degree (a partition multiplicity law); the asymptotics of spectral counting is given by the Hardy–Ramanujan partition asymptotics (an H-R asymptotic law); and the second-order correction is the Rademacher series (an R-correction law).

The Chinese version says the same, clause by clause (“spectrum is the nonnegative integers; and the eigen-multiplicities are the partition numbers of the corresponding degree; ...”). The conjecture is a conjunction of four clauses (1)–(4). We refute

(2) *for every $n \geq 0$, the eigenvalue n of $N = -L$ has multiplicity exactly $p(n)$,*

where $p(n)$ is the number of partitions of n , so $p(0), p(1), p(2), \dots = 1, 1, 2, 3, 5, 7, \dots$. “The corresponding degree” of the eigenvalue n is n , because the eigenspace of n is the n -th Wiener chaos, which consists of Hermite polynomials of degree n . Shifted indexings are treated in Section 4. Refuting one clause of a conjunction refutes the conjunction, so clauses (1), (3) and (4) need not be interpreted. Clause (1) is in fact true.

The objects

Let (Ω, H, μ) be a Gaussian space with isonormal process W on a real Hilbert space H . Examples are classical Wiener space ($H = L^2[0, 1]$, so H is infinite-dimensional) and $\mathbb{R}^d$ with the standard Gaussian measure γ_d ($H = \mathbb{R}^d$, $W(e_i) = x_i$). The Ornstein–Uhlenbeck operator is $L = -\delta D$, where D is the Malliavin derivative and δ its adjoint. Let (e_i) be an orthonormal family in H . On a cylinder polynomial $F = f(W(e_0), \dots, W(e_{d-1}))$ with f a polynomial, L acts by

$$LF = (L_d f)(W(e_0), \dots, W(e_{d-1})), \quad L_d = \sum_{i < d} \left(\frac{\partial^2}{\partial x_i^2} - x_i \frac{\partial}{\partial x_i} \right) = \Delta - x \cdot \nabla. \quad (1)$$

(This is standard; see Nualart, *The Malliavin Calculus and Related Topics*, §1.4.) The map $f \mapsto f(W(e_0), \dots, W(e_{d-1}))$ is injective on polynomials, and it is an isometry from $L^2(\gamma_d)$ into $L^2(\mu)$ because the variables $W(e_i)$ are i.i.d. standard normal. Polynomials lie in the domain of the self-adjoint operator $N = -L$, and N extends $-L_d$ on them. Hence:

Lemma 1. *If $f_1, \dots, f_m$ are linearly independent polynomials with $L_d f_j = -n f_j$, then $F_j = f_j(W(e))$ are m linearly independent eigenvectors of N for the eigenvalue n . In particular $\text{mult}(n) \geq m$, where $\text{mult}(n) = \dim \ker(N - n)$.*

2 Polynomial computations

We identify a polynomial with its coefficient map $\beta \mapsto f_\beta$, where $\beta \in \mathbb{N}^{(\mathbb{N})}$ is a multi-index and $x^\beta = \prod_i x_i^{\beta_i}$. Then $(\partial_i f)_\beta = (\beta_i + 1) f_{\beta + e_i}$ and $(x_i f)_\beta = f_{\beta - e_i}$ (which is 0 if $\beta_i = 0$). So $(x_i \partial_i f)_\beta = \beta_i f_\beta$ and

$$(L_d f)_\beta = \sum_{i < d} \left((\beta_i + 1)(\beta_i + 2) f_{\beta + 2e_i} - \beta_i f_\beta \right). \quad (2)$$

These are exactly the definitions used in Lean (`deriv`, `mulX`, `L`, `L.apply`). As a sanity check, Lean also proves the canonical commutation relation $[\partial_i, x_i] = 1$.

Lemma 2 (squarefree monomials). *If $\alpha_i \leq 1$ for all i and $\alpha_i = 0$ for $i \geq d$, then $L_d x^\alpha = -|\alpha| x^\alpha$ with $|\alpha| = \sum_i \alpha_i$.*

Proof. $\partial_i^2 x^\alpha = 0$ since $\alpha_i \leq 1$, and $x_i \partial_i x^\alpha = \alpha_i x^\alpha$. Sum over $i < d$. □

In particular $L_d x_i = -x_i$ for $i < d$, and $L_d(x_0 x_1 \cdots x_{n-1}) = -n x_0 \cdots x_{n-1}$. Distinct monomials are linearly independent.

Theorem 3 ($d \geq 2$). *On $\mathbb{R}^d$ with $d \geq 2$, $\text{mult}(1) \geq d \geq 2 > 1 = p(1)$. More generally $\text{mult}(n) \geq \lfloor d/n \rfloor$ for $n \geq 1$.*

Proof. The monomials $x_0, \dots, x_{d-1}$ are independent eigenvectors for the eigenvalue 1 (Lemma 2, Lemma 1). For general n , the block monomials $b_j = x_{nj} x_{nj+1} \cdots x_{nj+n-1}$ with $0 \leq j < \lfloor d/n \rfloor$ are distinct squarefree monomials of degree n . □

Theorem 4 (Wiener space). *If $\dim H = \infty$ (for example on classical Wiener space), then every eigenvalue $n \geq 1$ of N has infinite multiplicity, so $\text{mult}(n) \neq p(n)$ for every $n \geq 1$.*

Proof. Choose an orthonormal sequence $e_0, e_1, \dots$ in H . For every m , the functionals $\prod_{k < n} W(e_{nj+k})$ with $j < m$ are m independent eigenvectors (Theorem 3 with $d = nm$, and Lemma 1). For $n = 1$ these are just $W(e_0), \dots, W(e_{m-1})$. □

Theorem 5 ($d = 1$). On $(\mathbb{R}, \gamma_1)$ every eigenvalue n has multiplicity exactly 1, with eigenspace $\mathbb{R}H_n$ (H_n the probabilists' Hermite polynomial). Hence $\text{mult}(n) = 1 < 2 \leq p(n)$ for every $n \geq 2$. In particular, for $n = 2$ the eigenvectors are exactly the multiples of $x^2 - 1$, while $p(2) = 2$.

Proof. Polynomial eigenvectors. Write $g_k = f_{(k)}$ for the coefficient of x^k . By (2), $L_1 f = -nf$ is equivalent to

$$(k+1)(k+2)g_{k+2} = (k-n)g_k \quad (k \geq 0). \quad (3)$$

Suppose g is finitely supported, satisfies (3), and $g_n = 0$. For $k \neq n$, (3) shows that $g_{k+2} = 0$ implies $g_k = 0$. Descending from the top of the support gives $g_k = 0$ for all $k > n$. Then $g_{n+1} = 0$ and $g_n = 0$, and descending from there gives $g_k = 0$ for $k < n$. So $g = 0$. If f and f' are two eigenvectors, then $f'_n f - f_n f'$ is an eigenvector with vanishing x^n -coefficient, hence zero. So any two eigenvectors are proportional, and the polynomial eigenspace has dimension ≤ 1 . It contains H_n , which satisfies $H''_n - xH'_n = -nH_n$; for example $L_1(x^2 - 1) = 2 - 2x^2$. All L^2 eigenvectors. The Hermite polynomials H_k form an orthogonal basis of $L^2(\gamma_1)$ and $NH_k = kH_k$. If $NF = nF$, then for each k , $k\langle F, H_k \rangle = \langle F, NH_k \rangle = \langle NF, H_k \rangle = n\langle F, H_k \rangle$ by self-adjointness. So $\langle F, H_k \rangle = 0$ for $k \neq n$, and $F \in \mathbb{R}H_n$. Finally, for $n \geq 2$ the partitions (n) and $(1, \dots, 1)$ are distinct, so $p(n) \geq 2$. $\square$

Theorem 6 (exact multiplicities). On $(\mathbb{R}^d, \gamma_d)$, $\text{mult}(n) = \binom{n+d-1}{d-1}$, the number of multi-indices $\alpha \in \mathbb{N}^d$ with $|\alpha| = n$.

Proof. The products $H_\alpha(x) = \prod_i H_{\alpha_i}(x_i)$ form an orthogonal basis of $L^2(\gamma_d)$, and $NH_\alpha = |\alpha|H_\alpha$ because L_d is a sum of one-variable operators. The argument of Theorem 5 then shows that the eigenspace of n is $\text{span}\{H_\alpha : |\alpha| = n\}$. $\square$

The table below is recomputed by `verify.py` as kernel dimensions of $L_d + n$ on polynomials of degree $\leq n$ (and, as a check, $\leq n+2$):

n	0	1	2	3	4	5
$p(n)$	1	1	2	3	5	7
$d = 1$	1	1	1	1	1	1
$d = 2$	1	2	3	4	5	6
$d = 3$	1	3	6	10	15	21
$d = 4$	1	4	10	20	35	56

Note the isolated coincidence $d = 2, n = 4$ ($5 = p(4)$). The law already fails there at $n = 1, 2, 3$.

3 Main theorem

Theorem 7. Clause (2) is false on every Gaussian space: on $\mathbb{R}^d$ for every $d \geq 0$, and on every Wiener space with $\dim H = \infty$. Hence Conjecture 00000008579 is false.

Proof. For $d = 0$ (constants only) there is no eigenvector for $n = 1$, but $p(1) = 1$. For $d = 1$ use Theorem 5 at $n = 2$, for $d \geq 2$ Theorem 3 at $n = 1$, and for $\dim H = \infty$ Theorem 4 at $n = 1$. In every case the second clause fails, so the conjunction fails. $\square$

4 Robustness: other readings

1. *Which Wiener space.* Classical Wiener space, any abstract Wiener space, or any Gaussian Hilbert space of infinite dimension: Theorem 4. Finite-dimensional Gaussian spaces: Theorems 3 and 5. The answer does not depend on the choice of H , apart from its dimension.
2. *Field and kind of multiplicity.* N is self-adjoint, so algebraic and geometric multiplicities agree. Complexifying does not change dimensions. Lean measures rank over $\mathbb{Z}$, which equals the dimension over $\mathbb{Q}$ (clear denominators) and over $\mathbb{R}$ for the rational polynomials used.
3. *Shifted index* ($\text{mult}(n) = p(n + s)$). For $d = 1$, $1 \neq p(n + s)$ as soon as $n + s \geq 2$. For $d \geq 2$ the reading fails by item 4, and `verify.py` finds an explicit failure for $s \in \{-1, 1, 2\}$ and all $d \leq 29$. On infinite-dimensional Wiener space the multiplicities are infinite.
4. *“For all large n ” / asymptotic readings.* For finite d , $\binom{n+d-1}{d-1}$ is a polynomial of degree $d - 1$ in n . Meanwhile $p(n) \geq \binom{\lfloor n/(d+1) \rfloor + d}{d}$: choose multiplicities $m_2, \dots, m_{d+1} \geq 0$ with $\sum m_i \leq \lfloor n/(d+1) \rfloor$, so that $\sum_i i m_i \leq n$, and fill up with parts 1. This lower bound is a polynomial of degree d , so $p(n) > \text{mult}(n)$ for all large n ; e.g. for $d = 1, 2, 3$, for all $n > 1, 4, 13$. The same comparison refutes clause (3) for finite d : the counting function $\#\{\text{eigenvalues} \leq \lambda\} = \binom{\lfloor \lambda \rfloor + d}{d}$ is a polynomial, not of Hardy–Ramanujan type. For $\dim H = \infty$ every multiplicity is infinite.
5. *Restricting to symmetric functionals.* If one keeps only eigenvectors invariant under permutations of the d coordinates, the multiplicity of n is the number of partitions of n into at most d parts. This equals $p(n)$ only for $n \leq d$, so for each fixed d it fails at $n = d + 1$ (checked in `verify.py` for $d \leq 3$). With infinitely many i.i.d. coordinates, the Hewitt–Savage 0–1 law makes every exchangeable L^2 functional a.s. constant, so the multiplicities are 0 for $n \geq 1$.
6. *Is there any reading that gives $p(n)$?* Yes, but only for a different operator. The second quantization $d\Gamma(A)$ of $A = \text{diag}(1, 2, 3, \dots)$, i.e. $\sum_k k(\partial_k^2 - x_k \partial_k)$ (an Ornstein–Uhlenbeck operator with drift $-A$, or a string-oscillator energy operator), has eigenvalue $\sum_k k \alpha_k$ on H_α . Its multiplicities are therefore $p(n)$ (`verify.py` checks this for $d = 4, 5$, $n \leq d$). The conjecture names *the* Ornstein–Uhlenbeck operator of Malliavin calculus, which is $d\Gamma(I) = -L$, the number operator. Its eigenspaces are the Wiener chaoses, and the “Hermite spectrum law” of clause (1) refers to exactly these Hermite eigenfunctions. We therefore regard the weighted operator as a different conjecture, not a reading of this one.

5 Lean formalization

The Lean 4.19.0 project in `lean4/` uses the core library only. Its definitions are as follows.

- `Poly := (Nat -> Nat) -> Int`, the coefficient map, and `mono, x i, H2 = x02 - 1`.
- The predicates `InVars`, `Bounded` and `IsPoly d` (finite support in $x_0, \dots, x_{d-1}$).
- `deriv, mulX`, and `L d f = fun b => isum d (fun i => deriv i (deriv i f) b - mulX i (deriv i f) b)`.
- `Eig d n f` (f is a polynomial with $Lf = -nf$), `LinIndep`, `MultGE d n m`, `Mult d n m` ($\geq m$ and not $\geq m + 1$), and the cylinder versions `EigCyl`, `MultGECyl`, `MultCyl`.
- `IsPartition`, `partitionsL, p`, with `mem_partitionsL` for all n .

The main theorems are as follows.

- `L_apply` (formula (2)) and `deriv_mulX_comm` ($[\partial_i, x_i] = 1$).
- `L_x0_x1` ($Lx_0 = -x_0, Lx_1 = -x_1$) and `linIndep_x`.
- `eig_H2` ($L(x^2 - 1) = -2(x^2 - 1)$) and `multGE.blocks` ($\text{mult}_d(n) \geq m$ when $nm \leq d$).
- `eig1_prop`: in one variable, any two eigenvectors for the same eigenvalue, of any degree, are proportional. Consequences: `mult1_le` ($\text{mult}_1(n) \leq 1$ for all n), `eig2_eq_H2` (every eigenvector for 2 is $f_2 \cdot (x^2 - 1)$), and `mult1_exact` ($\text{mult}_1(0) = \text{mult}_1(1) = \text{mult}_1(2) = 1$).
- `p-values` ($p(0..5) = 1, 1, 2, 3, 5, 7$) and `p_ge_two`.
- `partitionLaw_false` $d: \neg \text{PartitionLaw } d$ for every d ; `d1_fails_all` (for $d = 1$ the law fails at every $n \geq 2$); `mult_one_ge_two`.
- `L_eq_of_isPoly`: L is consistent across dimensions, so it is well defined on cylinder polynomials. `multCyl_infinite` (no finite multiplicity for any $n \geq 1$) and `partitionLawCyl_false`.
- `conjecture_00000008579_false` and `conjecture_00000008579_false_wiener`: $\neg(C_1 \wedge (2) \wedge C_3 \wedge C_4)$ for arbitrary propositions C_1, C_3, C_4 .

The project contains no `sorry`, no `native_decide` and no added axioms. `#print axioms` reports `propext`, `Classical.choice` and `Quot.sound`. Classical choice is used only to decide equality of multi-indices, which are functions, in the definition of monomials.

Limitations. Lean works on the polynomial core and does not formalize $L^2(\mu)$, the closure of L , or Hermite completeness. For $d \geq 2$ and for Wiener space the disproof needs only lower bounds, and Lemma 1 transfers polynomial eigenvectors to genuine eigenvectors. For $d = 1$ it needs the upper bound $\text{mult}(2) \leq 1$. Lean proves this bound for the polynomial eigenspace, and the extension to all of $L^2(\gamma_1)$ (Hermite completeness) is Theorem 5, proved here only. Existence of H_n for general n (so that $\text{mult}_1(n) = 1$ exactly) and the exact formula of Theorem 6 are proved in this report and checked numerically in `verify.py`. Lean proves $\text{mult}_1(n) = 1$ exactly only for $n \leq 2$, and for $d \geq 2$ it proves lower bounds only. The readings in Section 4, items 3–6, are covered by this report and `verify.py`, except for $d = 1$ at every $n \geq 2$, which is in Lean.

6 Independent check

`verify.py` (Python 3 standard library) uses a different method. It builds the matrix of L_d on the monomial basis and computes kernel dimensions of $L_d + n$ exactly over $\mathbb{Q}$. It checks the following.

- The table above for $d \leq 4$, $n \leq 5$, and that degree $\leq n + 2$ gives the same dimensions.
- $\text{mult}_d(1) = d$ for $d \leq 7$.
- L of the Hermite polynomials for $n \leq 12$.
- $p(n)$ by Euler’s recurrence and by enumeration.
- The shifted, eventual, symmetric and weighted-operator readings of Section 4.

All checks pass.